

Data Security

Master's in Cybersecurity

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Data Security

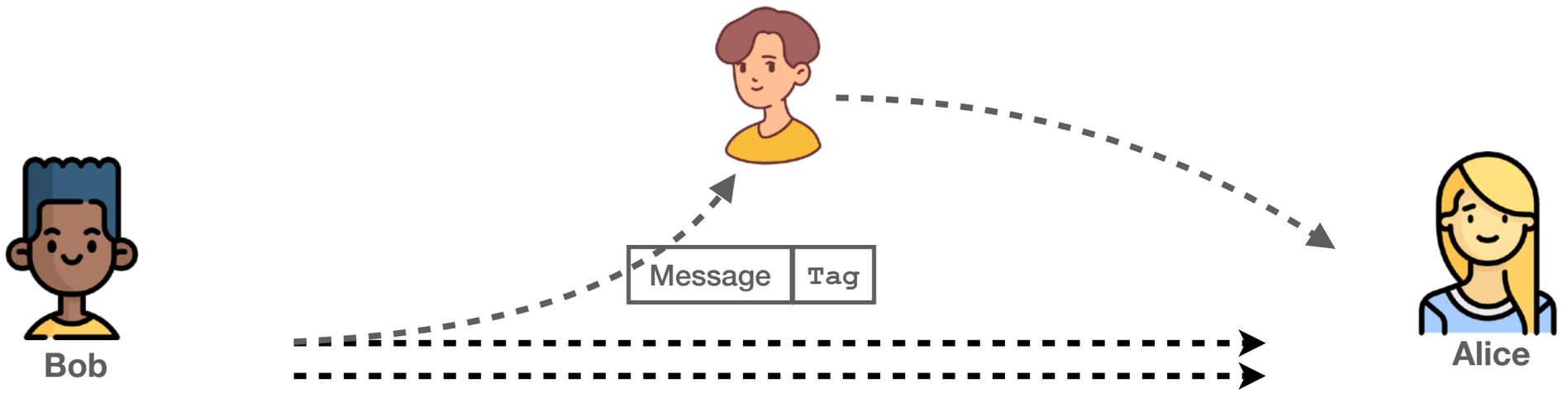
Objectives

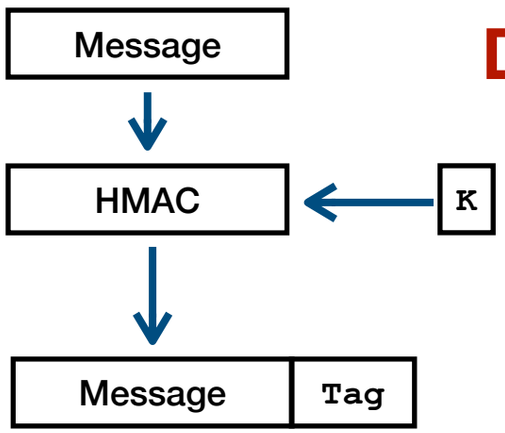
- Understand the role of digital signatures in message authentication
- Understand the properties and requirements of digital signature schemes
- Understand the relationship between digital signatures and the non-repudiation property
- Know the main approaches to digital signatures
- Understand the importance of a secure association between public keys and underlying entities

Recap

Message authentication

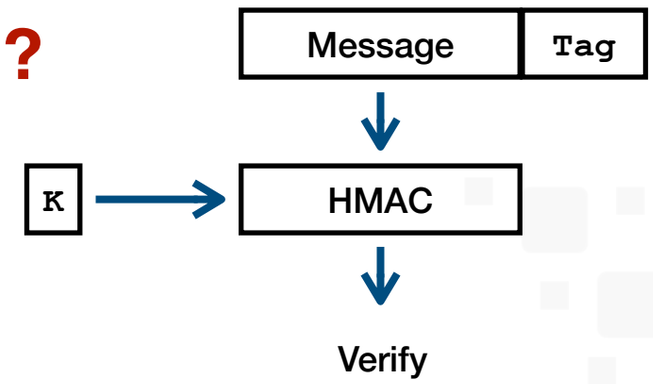
Which security properties are protected in this scheme?





Does it protect Bob from Alice?

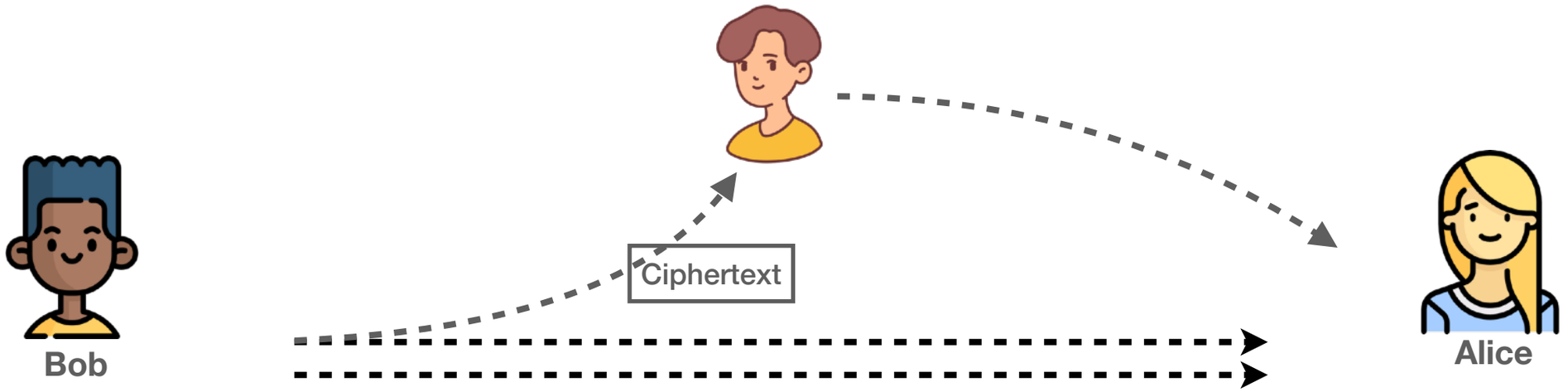


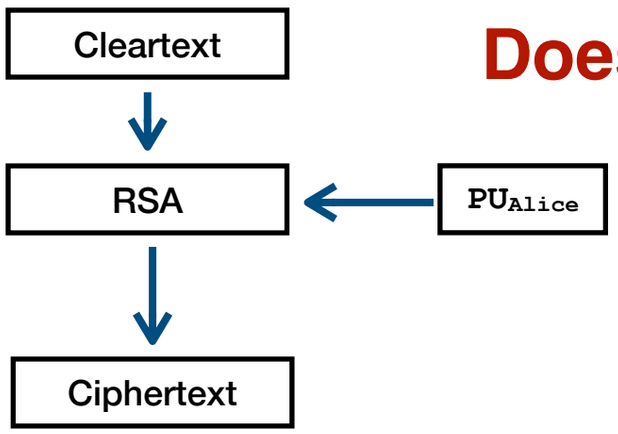


Recap

Public-key cryptography

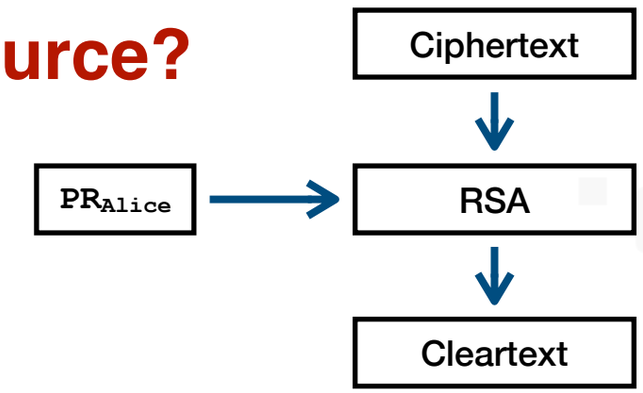
Which security properties are protected in this scheme?





Does it authenticate the source?





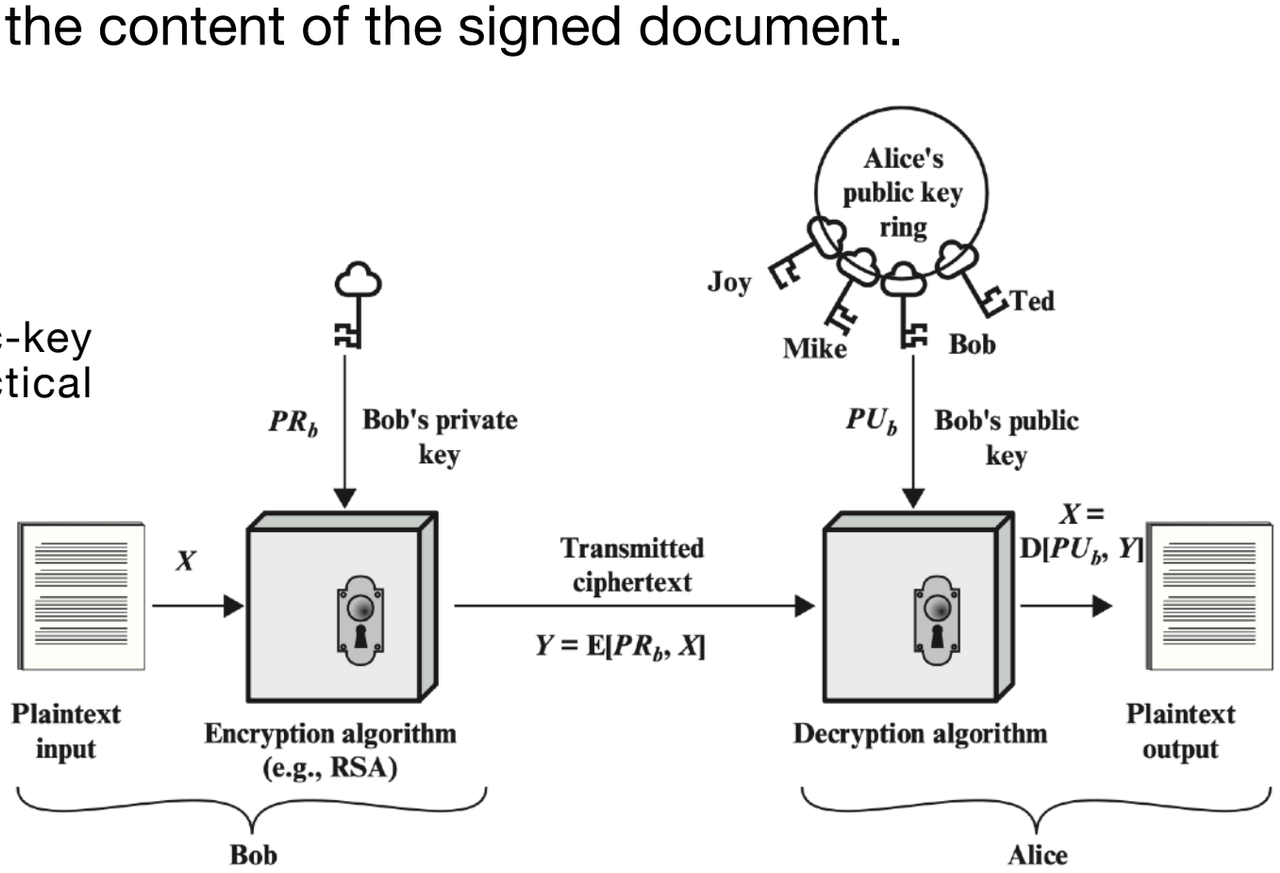
Public-key Cryptography

Digital signatures

A mechanism that uniquely associates a signer to a document and allows this association to be validated by third parties.

- It also guarantees the integrity of the signed document.
- One advantage over conventional signatures is that they are always different and dependent on the content of the signed document.

The overhead of public-key cryptography is a practical challenge.



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Public-key Cryptography

Digital signatures

Properties

- It must verify the author
  - the date and time of the signature
- It must authenticate the contents at the time of the signature
- It must be verifiable by third parties to resolve disputes

Public-key Cryptography

Digital signatures

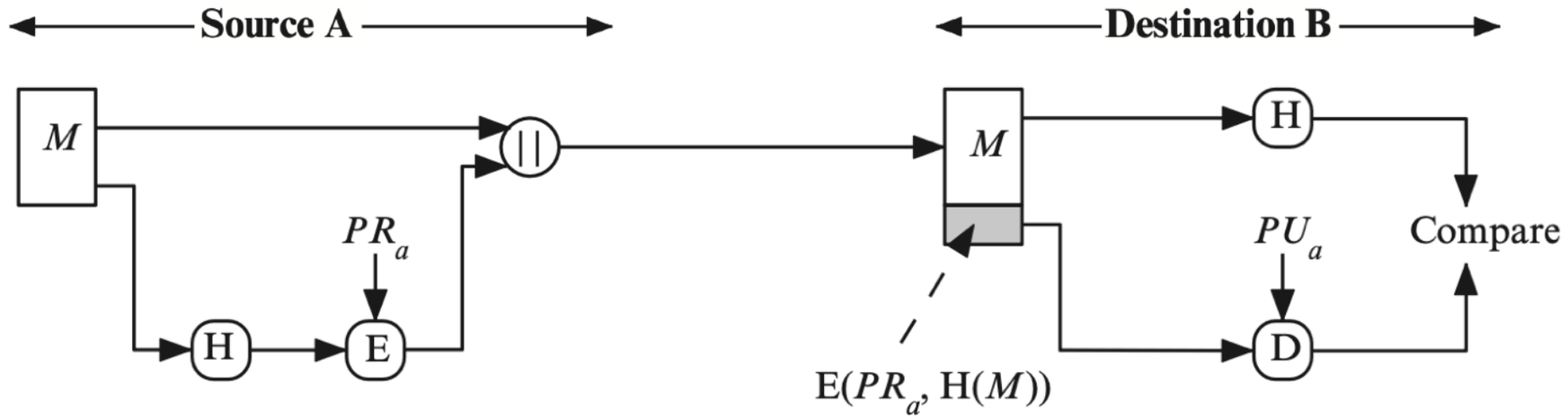
Requirements

- The signature must be a bit pattern that depends on the signed message.
- The signature must use data only known to the signer. Preventing forgery and denial.
- It must be relatively easy to produce the digital signature.
- It must be relatively easy to recognise and verify the digital signature.
- It must be computationally infeasible to forge a digital signature.
  - By constructing a new message for an existing signature.
  - By constructing a fraudulent signature for a given message.
- It must be practical to retain a copy of the digital signature.

Public-key Cryptography

Digital signatures

A generic approach based on cryptographic functions



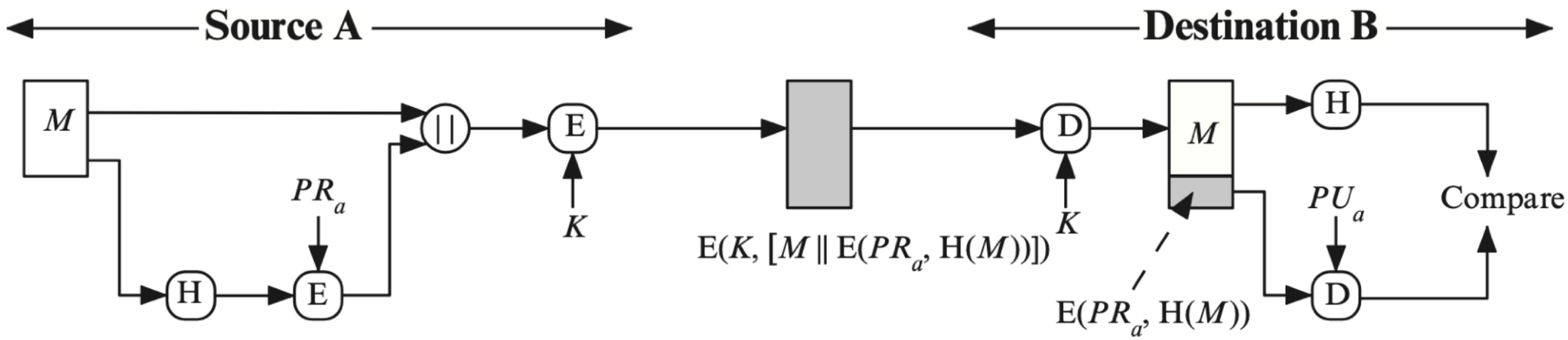
The hash code is encrypted using public-key encryption with the sender's private key.

- It provides authentication (signer authenticity + message integrity)
- It also provides non-repudiation since only the sender could have produced the hash code.

Public-key Cryptography

Digital signatures

A generic approach based on cryptographic functions



It adds confidentiality to the transmitted data.

- The message plus the private-key-encrypted hash code is encrypted using a symmetric key.

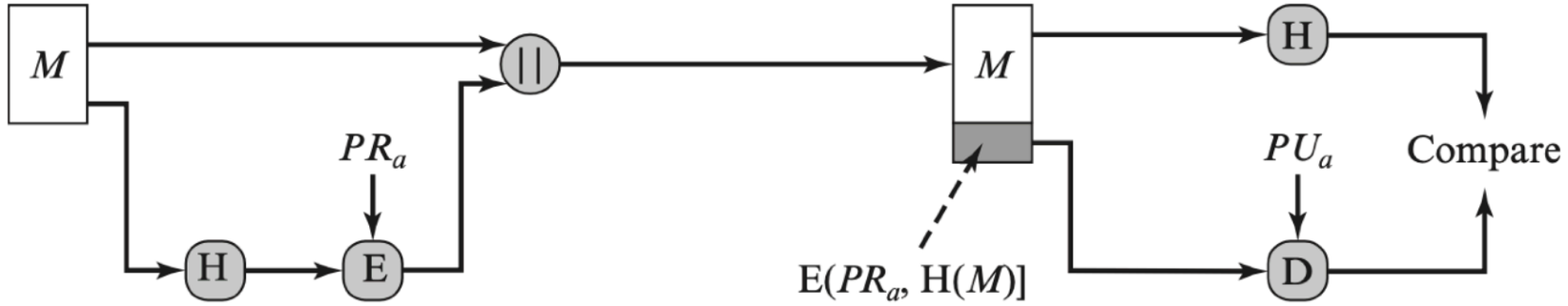
It requires a mechanism to share the symmetric key.

Public-key Cryptography

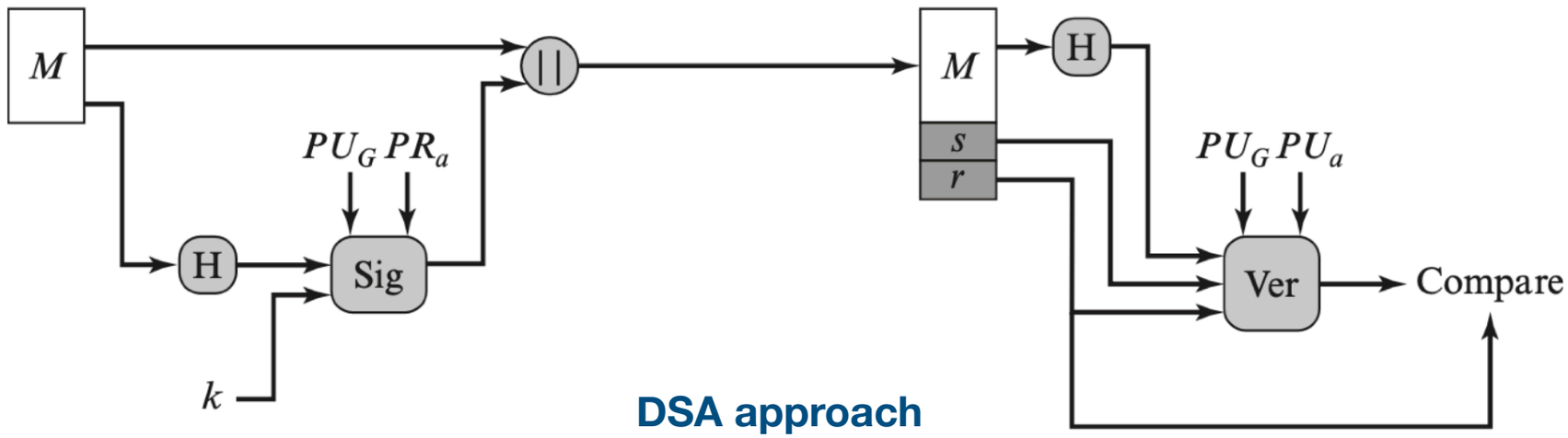
Digital signatures

Direct signature: a scheme that involves only the communicating parties (*signer* and *verifier*). It assumes the verifier knows the signer's public key.

Approaches



RSA approach



DSA approach

Public-key Cryptography

Digital signatures

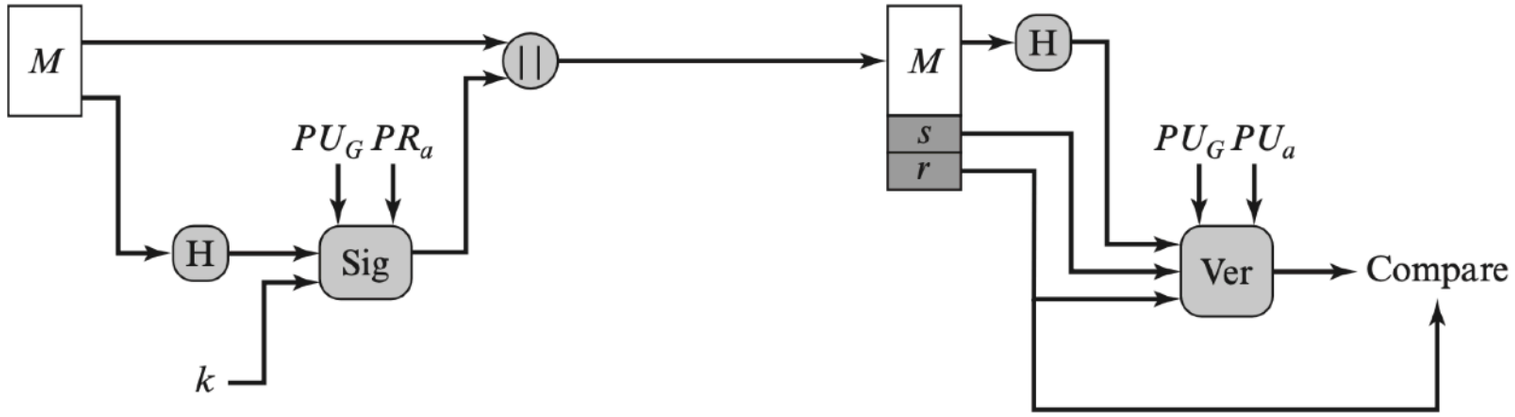
- *Digital Signature Standard* (*DSS*): The NIST standard for digital signature algorithm introduced in 1991,
  - Based on the ElGamal scheme.

- *Digital Signature Algorithm* (DSA)
  - Uses the Secure Hash Algorithm (SHA) to generate a digest of variable-size messages.
  - Designed to provide only the digital signature function
  - Designed to be more efficient than the RSA signature scheme.
    - Suitable for constrained devices, *e.g., smartcards.*
    - However, the verification process is comparatively heavier.

Public-key Cryptography

Digital signatures

DSA general scheme



- The signature function receives as input the message hash value and a random number `k` generated for a particular signature.
- The signature function also uses:
  - The signer private key (`PRa`).
  - Global public key (PU_G): a set of parameters known to a group of communicating principals.
- The resulting signature consists of two components, labelled as `s` and `r`.

Public-key Cryptography

Digital signatures

The DSA algorithm

Global Public-Key Components

p prime number where $2^{L-1} < p < 2^L$
for $512 \leq L \leq 1024$ and L a multiple of 64;
i.e., bit length L between 512 and 1024 bits
in increments of 64 bits

q prime divisor of $(p - 1)$, where $2^{N-1} < q < 2^N$
i.e., bit length of N bits

g $= h(p - 1)/q$ is an exponent mod p,
where h is any integer with $1 < h < (p - 1)$
such that $h^{(p-1)/q} \bmod p > 1$

User's Private Key

x random or pseudorandom integer with $0 < x < q$

User's Public Key

$y = g^x \bmod p$

User's Per-Message Secret Number

gdc(k,p-1)=1

k random or pseudorandom integer with $0 < k < q$

Signing

$r = (g^k \bmod p) \bmod q$

$s = [k^{-1} (H(M) + xr)] \bmod q$

Signature $= (r, s)$

Verifying

$w = (s')^{-1} \bmod q$

$u_1 = [H(M')w] \bmod q$

$u_2 = (r')w \bmod q$

$v = [(g^{u1} y^{u2}) \bmod p] \bmod q$

TEST: $v = r'$

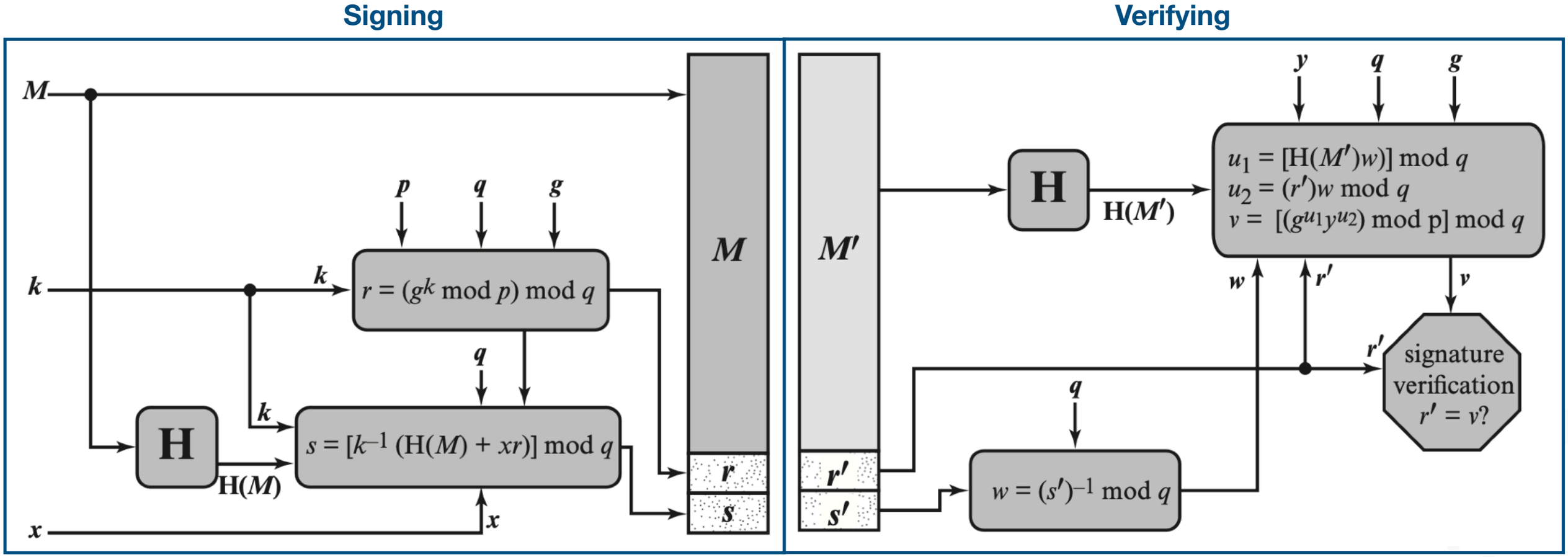
M = message to be signed
H(M) = hash of M using SHA-1
M', r', s' = received versions of M, r, s

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Public-key Cryptography

Digital signatures

The DSA algorithm



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Public-key Cryptography

Applications

Applications for the main public-key cryptosystems

Algorithm	Encryption/Decryption	Digital Signature	Key Exchange
RSA	Yes	Yes	Yes
Elliptic Curve	Yes	Yes	Yes
Diffie–Hellman	No	No	Yes
DSS	No	Yes	No

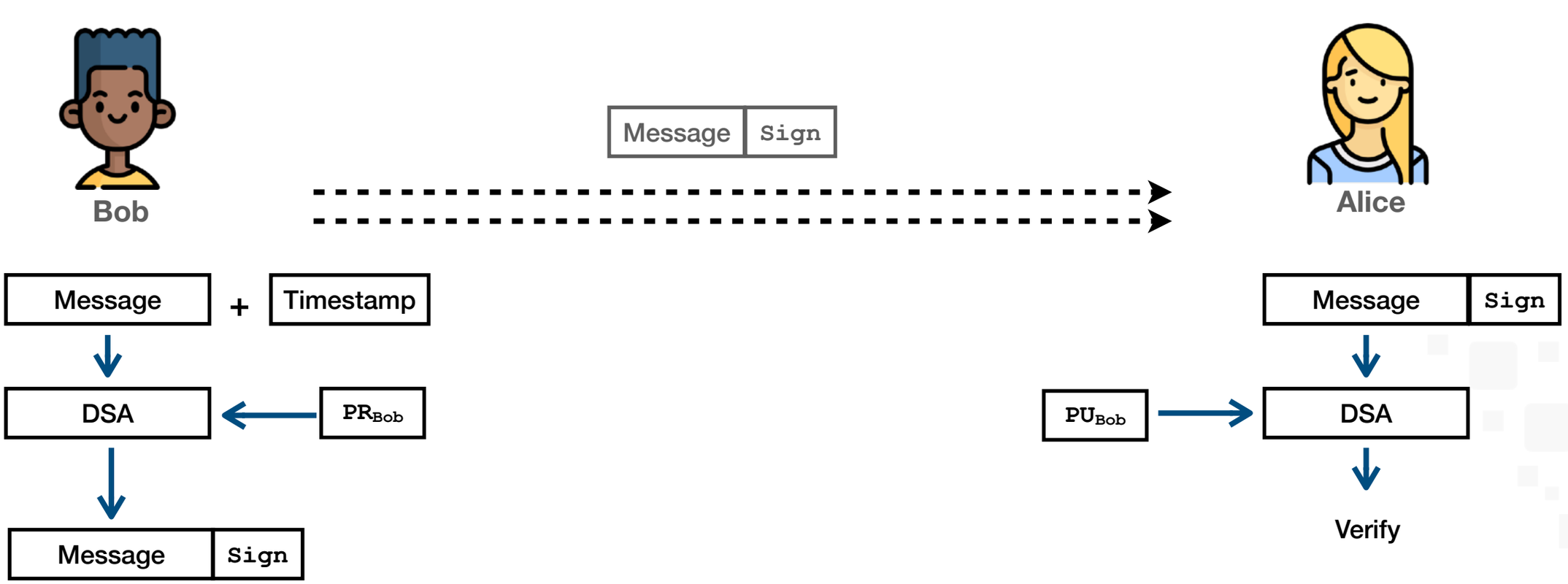
Public-key Cryptography

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New threats?

How can we protect against Bob forging the signature timestamp?

How can Alice securely obtain Bob's public key?





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